Amit Kumar Das

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Research Interests

Natural Language Processing, Data Science, Embodied AI and Human-Robot Interaction, Human-AI Collaboration, Intelligent User Interfaces, Large Language Models for Human-Centered Applications, Information Retrieval and Knowledge Discovery, AI for Accessibility and Inclusion, Explainable and Transparent AI, Data Visualization and Visual Analytics

Education

- **Stony Brook University** — Stony Brook, New York
  Ph.D. in Computer Science — *Jan 2022 - Aug 2026 (Expected)*
  - **Advisor**: Dr. Klaus Mueller
  - **Dissertation Focus**: Human-Centered AI for Data Visualization and Accessibility
  - **Relevant Coursework**: Visualization, Machine Learning, Theory of Database Systems, Logic in Computer Science, Fundamentals of Computer Networks
- **University of Dhaka** — Dhaka, Bangladesh
  Master of Science in Computer Science — *Jan 2014 - Jun 2015*
  - **Thesis**: A QoS and Profit-Aware Local and Global Cloud Confederation Model
  - **Relevant Coursework**: Image Processing, Advanced Logic Design, Network Performance Analysis, Project Management, Cloud Computing
- **University of Dhaka** — Dhaka, Bangladesh
  Bachelor of Science in Computer Science — *Feb 2009 - Oct 2013*
  - **Relevant Coursework**: Artificial Intelligence, Distributed Systems, Computer Graphics, Compiler Design and Construction, Software Engineering

Academic Appointments

- **Stony Brook University** — Stony Brook, New York
  Research Assistant & Teaching Assistant — *Jan 2022 - Present*
  - **Research**: Conduct research in Natural Language Processing, Human-AI Interaction, Visual Analytics, and Large Language Models for human-centered applications. Lead multiple interdisciplinary projects from conception to publication in top-tier venues (IEEE VIS, CHI).
  - **Teaching Assistant**: Serve as TA for Foundations of Computer Science (CSE 115) and Visualization (CSE 564) courses. Prepare and deliver recitation sessions, grade assignments, hold office hours, and provide one-on-one student support.
  - **Technical Development**: Build interactive dashboards and visualization tools for data analysis. Develop machine learning models and conduct user studies to validate research hypotheses.

Publications

Citations: 2,058 — h-index: 28 — i10-index: 43 (Google Scholar)

Papers Under Review

1. **A. K. Das**, K. Mueller, "GraphWhisper: LLM-Based Conversational Chart Question Answering for Blind and Low-Vision Users," Submitted to *Proc. CHI Conference on Human Factors in Computing Systems (CHI 2026)*, Under Review.

2. **A. K. Das**, C. X. Bearfield, K. Mueller, "Can LLM Understand Personal Perspectives? Leveraging Large Language Models for Personalized Public Messaging," Submitted to *Proc. CHI Conference on Human Factors in Computing Systems (CHI 2026)*, Under Review.

3. **A. K. Das**, K. Mueller, "VisNoob: An Interactive Dashboard to Assist Non-experts Understand and Fix Misleading Data Visualizations," Submitted to *IEEE Transactions on Visualization & Computer Graphics (TVCG)*, Under Review.

Peer-Reviewed Conference Papers (Full Papers)

2025

C1. **A. K. Das**, M. Tarun, K. Mueller, "Charts-of-Thought: Enhancing LLM Visualization Literacy Through Structured Data Extraction," In Proceedings of the IEEE Visualization Conference (IEEE VIS), 2025. (Accepted)

C2. **A. K. Das**, K. Mueller, "MisVisFix: An Interactive Dashboard for Detecting, Explaining, and Correcting Misleading Visualizations using Large Language Models," In Proceedings of the IEEE Visualization Conference (IEEE VIS), 2025. (Accepted)

C3. **A. K. Das**, C. X. Bearfield, K. Mueller, "Leveraging Large Language Models for Personalized Public Messaging," In Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA), 2025, pp. 1–7.

2022

C4. H. Huda, M. A. I. Fahad, M. Islam, **A. K. Das**, "Bangla handwritten character and digit recognition using deep convolutional neural network on augmented dataset and its applications," In Proceedings of the 16th International Conference on Ubiquitous Information Management and Communication (IMCOM), 2022, pp. 1–7.

C5. I. J. Yra, M. H. Pranto, **A. K. Das**, "HELLO: An android-based mobile application for autism individuals to improve socio-communicational learnability in Bangladesh," In Proceedings of the 16th International Conference on Ubiquitous Information Management and Communication (IMCOM), 2022, pp. 1–7.

C6. S. Siddique, A. I. Hridoy, S. A. Khushbu, **A. K. Das**, "CVD: An improved approach of software vulnerability detection for object oriented programming languages using deep learning," In Proceedings of the Future Technologies Conference, 2022, pp. 145–164.

2021

C7. N. Ahmad, T. Sharmin, J. Akter, R. A. Tuhin, **A. K. Das**, "Behavioral issues of children in terms of internet usages time in Bangladesh," In Proceedings of the 15th International Conference on Ubiquitous Information Management and Communication (IMCOM), 2021, pp. 1–6.

C8. M. T. Hossain, M. W. Hasan, **A. K. Das**, "Bangla handwritten word recognition system using convolutional neural network," In Proceedings of the 15th International Conference on Ubiquitous Information Management and Communication (IMCOM), 2021, pp. 1–8.

C9. J. F. Sorna, M. M. U. Rashel, R. A. Tuhin, **A. K. Das**, "CNN based Security issues of the Drivers of Ridesharing Services in Bangladesh," In Proceedings of the 6th International Conference on Inventive Computation Technologies (ICICT), 2021, pp. 1273–1278.

C10. M. Rahman, S. Das, M. Z. Tazim, M. Rana, R. A. Tuhin, **A. K. Das**, "State of the Art of ICT based Telemedicine and E-health Services in Bangladesh," In Proceedings of the 6th International Conference on Inventive Computation Technologies (ICICT), 2021, pp. 1266–1272.

2020

C11. R. H. Shaon, M. A. Islam, M. S. H. Khan, **A. K. Das**, "A Digital Platform Design for Supply Chain of Existing Fish Market in Bangladesh," In Proceedings of International Joint Conference on Computational Intelligence (IJCCI), 2020, pp. 497–513.

2019

C12. O. F. Rakib, S. Akter, M. A. Khan, **A. K. Das**, K. M. Habibullah, "Bangla word prediction and sentence completion using GRU: an extended version of RNN on N-gram language model," In Proceedings of the International Conference on Sustainable Technologies for Industry 4.0 (STI), 2019, pp. 1–6.

C13. S. T. Cynthia, K. M. S. Hossain, M. N. Hasan, M. Asaduzzaman, **A. K. Das**, "Automated detection of plant diseases using image processing and faster R-CNN algorithm," In Proceedings of the International Conference on Sustainable Technologies for Industry 4.0 (STI), 2019, pp. 1–5.

C14. S. Paul, J. I. Joy, S. Sarker, A. Shakib, S. Ahmed, **A. K. Das**, "An unorthodox way of farming without intermediaries through blockchain," In Proceedings of the International Conference on Sustainable Technologies for Industry 4.0 (STI), 2019, pp. 1–6.

C15. S. T. Cynthia, M. Majumder, A. Tabassum, N. N. Khanom, R. A. Tuhin, **A. K. Das**, "Security concerns of ridesharing services in Bangladesh," In Proceedings of the 2nd International Conference on Applied Information Technology and Innovation (ICAITI), 2019, pp. 44–50.

C16. A. Hossain, **A. K. Das**, T. N. Mim, J. Hoque, R. A. Tuhin, "Software piracy: Factors and profiling," In Proceedings of the 2nd International Conference on Applied Information Technology and Innovation (ICAITI), 2019, pp. 213–219.

C17. M. Hassan, A. Saha, N. Saba, R. A. Tuhin, **A. K. Das**, "Impact of social media on socialization of university students (A Study on East West University's Undergraduate Students)," In Proceedings of the 2nd International Conference on Applied Information Technology and Innovation (ICAITI), 2019, pp. 173–178.

C18. S. S. Hassan, F. Nihar, M. Rahman, M. W. Razu, R. A. Tuhin, **A. K. Das**, "A Behavioral Model of Music piracy in Bangladesh: Factors influencing music piracy," In Proceedings of the 2nd International Conference on Applied Information Technology and Innovation (ICAITI), 2019, pp. 197–202.

C19. S. Sultana, M. H. Ullah, S. A. Nisher, M. S. Iqbal, M. R. A. Tuhin, **A. K. Das**, "Digital Governance for Pension Withdrawal System in Bangladesh," In Proceedings of the 2nd International Conference on Applied Information Technology and Innovation (ICAITI), 2019, pp. 185–189.

C20. M. S. H. Khan, P. Roy, F. Khanam, F. H. Hera, **A. K. Das**, "An efficient resource allocation mechanism for time-sensitive data in dew computing," In Proceedings of the International Conference of Artificial Intelligence and Information Technology (ICAIIT), 2019, pp. 506–510.

C21. M. M. Hossain, M. F. Labib, A. S. Rifat, **A. K. Das**, M. Mukta, "Auto-correction of english to bengali transliteration system using levenshtein distance," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C22. E. Biswas, **A. K. Das**, "Symptom-based disease detection system in bengali using convolution neural network," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C23. S. Paul, J. I. Joy, S. Sarker, A. Shakib, S. Ahmed, **A. K. Das**, "Fake news detection in social media using blockchain," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C24. J. B. Billa, A. Nawar, M. M. H. Shakil, **A. K. Das**, "PassMan: A new approach of password generation and management without storing," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C25. E. A. Emon, S. Rahman, J. Banarjee, **A. K. Das**, T. Mittra, "A deep learning approach to detect abusive bengali text," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C26. M. F. Labib, A. S. Rifat, M. M. Hossain, **A. K. Das**, F. Nawrine, "Road accident analysis and prediction of accident severity by using machine learning in Bangladesh," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C27. M. D. Drovo, M. Chowdhury, S. I. Uday, **A. K. Das**, "Named entity recognition in Bengali text using merged hidden Markov model and rule base approach," In Proceedings of the 7th International Conference on Smart Computing & Communications (ICSCC), 2019, pp. 1–5.

C28. J. Q. Yuki, M. M. Q. Sakib, Z. Zamal, K. M. Habibullah, **A. K. Das**, "Predicting Crime Using Time and Location Data," In Proceedings of the 7th International Conference on Computer and Communications Management (ICCM), 2019, pp. 124–128.

C29. S. Islam, S. I. A. Khan, M. M. Abedin, K. M. Habibullah, **A. K. Das**, "Bird Species Classification from an Image Using VGG-16 Network," In Proceedings of the 7th International Conference on Computer and Communications Management (ICCM), 2019, pp. 38–42.

C30. R. A. Tuhin, B. K. Paul, F. Nawrine, M. Akter, **A. K. Das**, "An automated system of sentiment analysis from bangla text using supervised learning techniques," In Proceedings of the 4th IEEE International Conference on Computer and Communication Systems (ICCCS), 2019, pp. 360–364.

C31. K. M. Habibullah, A. Alam, S. Saha, **A. K. Das**, "A Driver-Centric Carpooling: Optimal Route-Finding Model Using Heuristic Multi-Objective Search," In Proceedings of the 4th IEEE International Conference on Computer and Communication Systems (ICCCS), 2019, pp. 735–739.

C32. J. Islam, M. Mubassira, M. R. Islam, **A. K. Das**, "A speech recognition system for Bengali language using recurrent neural network," In Proceedings of the 4th IEEE International Conference on Computer and Communication Systems (ICCCS), 2019, pp. 73–76.

C33. **A. K. Das**, A. Ashrafi, M. Ahmmad, "Joint cognition of both human and machine for predicting criminal punishment in judicial system," In Proceedings of the 4th IEEE International Conference on Computer and Communication Systems (ICCCS), 2019, pp. 36–40.

C34. M. M. Shahriar, M. S. Iqubal, S. Mitra, **A. K. Das**, "A deep learning approach to predict malnutrition status of 0-59 month's older children in Bangladesh," In Proceedings of the IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT), 2019, pp. 145–149.

C35. M. Mubassira, **A. K. Das**, "The Impact of University Students' Smartphone use and academic performance in Bangladesh: a quantitative study," In Proceedings of the International Conference on Ubiquitous Information Management and Communication, 2019, pp. 734–748.

C36. N. Jahan, U. S. Fariha, M. R. Ananna, **A. K. Das**, "Pregnant women's condition and awareness about mood swings: A survey study in Bangladesh," In Proceedings of the International Conference on Ubiquitous Information Management and Communication, 2019, pp. 749–760.

2017

C37. M. R. Ullah, M. A. R. Bhuiyan, **A. K. Das**, "IHEMHA: Interactive healthcare system design with emotion computing and medical history analysis," In Proceedings of the 6th International Conference on Informatics, Electronics and Vision & 7th International Symposium in Computational Medical and Health Technology (ICIEV-ISCMHT), 2017, pp. 1–8.

C38. M. A. Al Mamun, J. A. Puspo, **A. K. Das**, "An Intelligent Smartphone Based Approach Using IoT for Ensuring Safe Driving," In Proceedings of the International Conference on Electrical Engineering and Computer Science (ICECOS), 2017, pp. 217–223.

C39. A. Tashnim, S. Nowshin, F. Akter, **A. K. Das**, "Interactive interface design for learning numeracy and calculation for children with autism," In Proceedings of the 9th International Conference on Information Technology and Electrical Engineering (ICITEE), 2017.

C40. F. T. Zohora, M. R. R. Khan, M. F. R. Bhuiyan, **A. K. Das**, "Enhancing the capabilities of IoT based fog and cloud infrastructures for time sensitive events," In Proceedings of the International Conference on Electrical Engineering and Computer Science (ICECOS), 2017, pp. 224–230.

C41. M. Akter, F. T. Zohra, **A. K. Das**, "Q-MAC: QoS and mobility aware optimal resource allocation for dynamic application offloading in mobile cloud computing," In Proceedings of the International Conference on Electrical, Computer and Communication Engineering (ECCE), 2017, pp. 803–808.

C42. M. S. Hoque, N. Shawkat, A. A. Chowdhury, **A. K. Das**, "Triple-E: Energy Efficient Extended Pool Management Scheme in Cloud Data Centers," In Proceedings of the International Conference on Innovations in Power and Advanced Computing Technologies (i-PACT), 2017.

2015

C43. T. Adhikary, **A. K. Das**, M. A. Razzaque, M. E. H. Chowdhury, S. Parvin, "Test implementation of a sensor device for measuring soil macronutrients," In Proceedings of the International Conference on Networking Systems and Security (NSysS), 2015, pp. 1–8.

2014

C44. **A. K. Das**, T. Adhikary, M. A. Razzaque, E. J. Cho, C. S. Hong, "A QoS and profit aware cloud confederation model for IaaS service providers," In Proceedings of the 8th International Conference on Ubiquitous Information Management and Communication (ICUIMC), 2014, pp. 1–7.

2013

C45. **A. K. Das**, T. Adhikary, M. A. Razzaque, C. S. Hong, "An intelligent approach for virtual machine and QoS provisioning in cloud computing," In Proceedings of the International Conference on Information Networking (ICOIN), 2013, pp. 462–467.

C46. T. Adhikary, **A. K. Das**, M. A. Razzaque, A. M. J. Sarkar, "Energy-efficient scheduling algorithms for data center resources in cloud computing," In Proceedings of the IEEE 10th International Conference on High Performance Computing and Communications & IEEE International Conference on Embedded and Ubiquitous Computing (HPCC-EUC), 2013, pp. 1715–1720.

2012

C47. T. Adhikary, **A. K. Das**, M. A. Razzaque, M. O. Rahman, C. S. Hong, "A distributed wake-up scheduling algorithm for base stations in green cellular networks," In Proceedings of the 6th International Conference on Ubiquitous Information Management and Communication (ICUIMC), 2012, pp. 1–7.

Journal Articles

2023

J1. R. Haque, N. Islam, M. Tasneem, **A. K. Das**, "Multi-class sentiment classification on Bengali social media comments using machine learning," International Journal of Cognitive Computing in Engineering, vol. 4, pp. 21–35, 2023. Elsevier.

2022

J2. M. Rahman, M. R. A. Talukder, L. A. Setu, **A. K. Das**, "A dynamic strategy for classifying sentiment from Bengali text by utilizing Word2vector model," Journal of Information Technology Research (JITR), vol. 15, no. 1, pp. 1–17, 2022. IGI Global.

2021

J3. A. Mohammad, M. Z. Utso, S. B. Habib, **A. K. Das**, "Predicting Retinal Diseases using Efficient Image Processing and Convolutional Neural Network (CNN)," Journal of Engineering Advancements, vol. 2, no. 04, pp. 221–227, 2021.

J4. A. Hossain, M. S. Hasan, M. M. Asif, **A. K. Das**, "Performance analysis on Bangla handwritten digit recognition using CNN and transfer learning," International Journal of Advanced Networking and Applications, vol. 13, no. 1, pp. 4809–4815, 2021. Eswar Publications.

J5. M. S. S. Khan, S. R. Rafa, **A. K. Das**, "Sentiment analysis on bengali facebook comments to predict fan's emotions towards a celebrity," Journal of Engineering Advancements, vol. 2, no. 03, pp. 118–124, 2021.

J6. M. Mubassira, **A. K. Das**, "Implementation of Recurrent Neural Network with Language Model for Automatic Articulation Identification System in Bangla," International Journal of Advanced Networking and Applications, vol. 12, no. 6, pp. 4800–4808, 2021. Eswar Publications.

J7. **A. K. Das**, A. Al Asif, A. Paul, M. N. Hossain, "Bangla hate speech detection on social media using attention-based recurrent neural network," Journal of Intelligent Systems, vol. 30, no. 1, pp. 578–591, 2021. De Gruyter.

J8. T. F. Mumu, I. J. Munni, **A. K. Das**, "Depressed people detection from bangla social media status using lstm and cnn approach," Journal of Engineering Advancements, vol. 2, no. 01, pp. 41–47, 2021.

J9. F. Nihar, N. N. Khanom, S. S. Hassan, **A. K. Das**, "Plant disease detection through the implementation of diversified and modified neural network algorithms," Journal of Engineering Advancements, vol. 2, no. 01, pp. 48–57, 2021.

2019

J10. M. A. R. Bhuiyan, M. R. Ullah, **A. K. Das**, "iHealthcare: Predictive Model Analysis Concerning Big Data Applications for Interactive Healthcare Systems," Applied Sciences, vol. 9, no. 16, p. 3365, 2019. MDPI.

2017

J11. **A. K. Das**, T. Adhikary, M. A. Razzaque, M. Alrubaian, M. M. Hassan, Z. Uddin, S. Biao, "Big media healthcare data processing in cloud: a collaborative resource management perspective," Cluster Computing, pp. 1–16, 2017. Springer.

J12. T. Adhikary, **A. K. Das**, M. A. Razzaque, M. Alrubaian, M. M. Hassan, A. Alamri, "Quality of service aware cloud resource provisioning for social multimedia services and applications," Multimedia Tools and Applications, vol. 76, no. 12, pp. 14485–14509, 2017. Springer.

2016

J13. T. Adhikary, **A. K. Das**, M. A. Razzaque, A. Almogren, M. Alrubaian, M. M. Hassan, "Quality of service aware reliable task scheduling in vehicular cloud computing," Mobile Networks and Applications, vol. 21, no. 3, pp. 482–493, 2016. Springer.

Research Experience & Projects

Large Language Models and Natural Language Processing *Jan 2024 - Present*

- Built framework combining LLMs with conjoint analysis for personalized public messaging, tested with 2,040 human participants, achieving 99.18% cost reduction
- Created interactive web dashboard enabling stakeholders to explore message effectiveness across personality types with real-time optimization
- Developed Charts-of-Thought prompting strategy achieving Claude-3.7-sonnet score of 50.17 on Visualization Literacy Assessment Test (baseline: 28.82)
- Built comprehensive dashboard for detecting and correcting misleading visualizations with 96% accuracy across 74 known issue types
- Extensive experience with transformer architectures, attention mechanisms, and state-of-the-art NLP techniques including BERT, GPT, and custom language models

Data Science and Knowledge Discovery *Jun 2023 - Present*

- Developed machine learning pipelines for data extraction and analysis from complex visualization formats
- Applied data mining techniques for pattern recognition in large-scale datasets
- Conducted statistical analysis and A/B testing for system evaluation with thousands of participants
- Built predictive models combining traditional ML algorithms with deep learning approaches

Embodied AI and Human-Robot Interaction *Jan 2022 - Present*

- Research on conversational AI agents for accessibility applications, enabling natural language interaction with data systems
- Developed intelligent interfaces that adapt to diverse user capabilities, particularly for blind and low-vision users
- Designed multi-modal interaction paradigms combining voice, text, and structured output
- Conducted extensive user studies evaluating human-AI collaboration in real-world scenarios

Personalized Communication Systems *Jun 2023 - Present*

- Developed full-stack chat application integrating Big Five personality traits with color-coded word emphasis using GPT-4 API
- Implemented real-time architecture using Node.js, Socket.IO, and Redis achieving sub-300ms latency
- Conducted six-week user study with 24 participants showing 41.6% improvement in emotional engagement
- Applied information retrieval techniques for personalized content recommendation and delivery

Teaching Experience

Stony Brook University Stony Brook, New York
Teaching Assistant *Jan 2022 - Present*

- **CSE 564: Visualization** (Graduate, Spring 2024): Conducted weekly recitation sessions, graded assignments and projects, held regular office hours
- **CSE 115: Foundations of Computer Science** (Undergraduate, Spring 2022): Led programming labs, provided individual student mentoring, graded exams and homework
- Received Best TA Award for excellence in teaching and student support

East West University Dhaka, Bangladesh
Senior Lecturer *May 2016 - Dec 2021*

- **Artificial Intelligence** (Undergraduate, 10 semesters, 40 students/section): Taught AI fundamentals including search algorithms, knowledge representation, machine learning basics, and intelligent agents
- **Machine Learning** (Undergraduate, 6 semesters, 35 students/section): Covered supervised/unsupervised learning, neural networks, deep learning, and practical implementations using Python
- **Natural Language Processing** (Undergraduate/Graduate): Taught NLP fundamentals, language models, text classification, and sequence-to-sequence models
- **C Programming** (Undergraduate, 6 semesters, 35 students/section): Introduced programming fundamentals, data structures, and algorithm design

- **Java Programming** (Undergraduate, 6 semesters, 30 students/section): Taught object-oriented programming concepts, GUI development, and software design patterns
- **Data Structures** (Undergraduate, 4 semesters, 30 students/section): Covered arrays, linked lists, trees, graphs, and complexity analysis
- **Algorithms** (Undergraduate, 4 semesters, 35 students/section): Taught algorithm design techniques, dynamic programming, graph algorithms, and computational complexity
- **Additional Courses**: Numerical Methods (2 semesters), Object Oriented Programming (2 semesters), Web Technology (2 semesters), Introduction to Computers (1 semester)
- Developed comprehensive course materials including lecture slides, programming assignments, and exams for all courses
- Integrated hands-on projects and real-world applications to enhance student learning
- Received Best Teaching Faculty awards multiple times with student evaluations averaging 4.7–5.0/5.0

Mentoring Experience

Graduate and Undergraduate Research Mentoring

- **Stony Brook University** (2022-Present): Mentor summer research students from CSIRE and SIMMONS programs, guiding them through full research lifecycle from problem formulation to implementation
- **East West University** (2016-2021): Mentored approximately 50 research groups (2-3 students each, 150+ total students) in computer science research projects. Nearly all groups published papers in peer-reviewed conferences or journals, with students as first authors and me as supervising faculty. Research areas included machine learning, natural language processing, computer vision, IoT, and healthcare informatics
- Served as programming contest trainer and coach, preparing students for national and international ACM ICPC competitions
- Guided numerous undergraduate and graduate thesis projects covering diverse topics in AI, machine learning, and software engineering
- Many former students now pursue Ph.D. programs in the United States, Canada, Australia, and Europe, or hold positions at major technology companies

Service

Conference & Journal Reviewing

- **CHI Conference on Human Factors in Computing Systems**: Papers reviewer (2025, 2026), Case Studies reviewer (2025)
- **IEEE Conference on Visualization (VIS)**: Papers reviewer (2025)
- **ACM Conference on Computer-Supported Cooperative Work (CSCW)**: Papers reviewer (2025, 2026)
- **ACM International Conference on Intelligent User Interfaces (IUI)**: Papers reviewer (2025)
- **IEEE Access Journal**: Regular reviewer
- **Future Generation Computer Systems (FGCS) - Elsevier**: Regular reviewer

- **International Conference on Sustainable Technologies for Industry (STI)**: Papers reviewer
- Additional reviewing for multiple IEEE and ACM conferences (24 peer review records verified on Publons)

Professional Experience

Robi Axiata Limited Dhaka, Bangladesh
Software Engineer *Jun 2015 - May 2016*

- Led development of mobile applications for digital services with integrated backend systems
- Designed and implemented RESTful APIs for mobile application integration
- Managed technical implementation of digital service projects coordinating with cross-functional teams

Green Networking Research Group, University of Dhaka Dhaka, Bangladesh
Research Assistant *Jan 2012 - Jun 2015*

- Led Destruction Management System using Sensor Data Cloud with ISIF Asia funding
- Contributed to Precision Agriculture Information System for Bangladesh under government funding
- Developed Real-Time Detection system for Crop Diseases using Image Sensor Network Technology
- Participated in SWAPNO project creating Online Training Framework for underserved populations

Grants & Fellowships

- **NSF Research Traineeship (BIAS-NRT) Fellowship** - Stony Brook University, selected for interdisciplinary research on bias in data, humans, and institutions 2024-Present
- **ISIF Asia Award** - Project "Shohoj Sonchoy-Easy Savings" for financial technology solutions 2014
- **ICT Ministry Innovation Fund** - Project "SWAPNO" for online training framework development 2015-2016
- **ICT Ministry Fellowship** - Awarded for excellence in Master's thesis research 2014-2015

Awards & Honors

- Two papers accepted to IEEE VIS 2025 (premier visualization conference, 23% acceptance rate)
- Paper accepted to CHI 2025 Extended Abstracts (premier HCI conference)
- Best TA Award, Stony Brook University
- Best Teaching Faculty awards (multiple times), East West University
- High-impact research: 2,058 citations, h-index of 28, i10-index of 43 (Google Scholar)
- Publications in top-tier venues, including IEEE VIS, CHI, and leading journals

Technical Skills

Programming Languages: Python, Java, JavaScript, C/C++, SQL, R, MATLAB
AI/ML Frameworks: PyTorch, TensorFlow, Transformers, OpenCV, Pandas, NumPy, Scikit-learn, SpaCy, NLTK
Web Technologies: React, Node.js, D3.js, HTML/CSS, Socket.IO, Redis
Cloud & Tools: AWS, Google Cloud Platform, Docker, Git, Kubernetes
Databases: PostgreSQL, MySQL, MongoDB, Neo4j
Specializations: Large Language Models, Natural Language Processing, Data Science and Mining, Embodied AI, Human-AI Interaction, Information Retrieval, Machine Learning, Deep Learning, Computer Vision

Additional Information

Work Authorization: U.S. Green Card holder (authorized to work without sponsorship)
Languages: English (fluent), Bengali (native)
PhD Defense: Scheduled for Summer 2026, ready to start Fall 2026